The Data Detective’s Toolkit: Diagnose, Monitor, and Fix with Semantic Link Labs



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Day of Data PNW • November 7, 2026

HI, I'M JASON

THE DATA SHEPHERD

Microsoft® Most Valuable Professional

Honored to be recognized by Microsoft.



I HERD THE DATA

- User Group Organizer
- Nashville, Tennessee
- Fabric + Power BI
- Teaching + Speaking







MY FAVORITE MODEL

TARDIS

TOOLS & ALLIES

Power BI | Microsoft Fabric | DuckDB | Python | SQL | VS Code

Standing on the Shoulders of Giants



Grateful for the people who share their knowledge, encourage us, answer questions, and help the rest of us learn faster.

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All slides, resources, and demo files available:

The Data Shepherd

slides.datashepherd.dev



Sound Familiar?

"IT WAS FINE ON FRIDAY"

Follow the Evidence

Meet the Investigators

YOU ARE HERE

Meet the Investigators

The Lab

Model Investigation

Best Practice Analyzer

Broken-Reference Repair

Refresh Manager

VertiPaq Analyzer

Report Analysis

Case Closed

Meet the investigators



Semantic Link

The foundation

Semantic Link

Microsoft’s first-party Fabric Python library

- Runs in Microsoft Fabric notebooks
- Preinstalled in the default runtime
- Query model data and inspect metadata

Meet the investigators



Semantic Link

The foundation

Semantic Link Labs

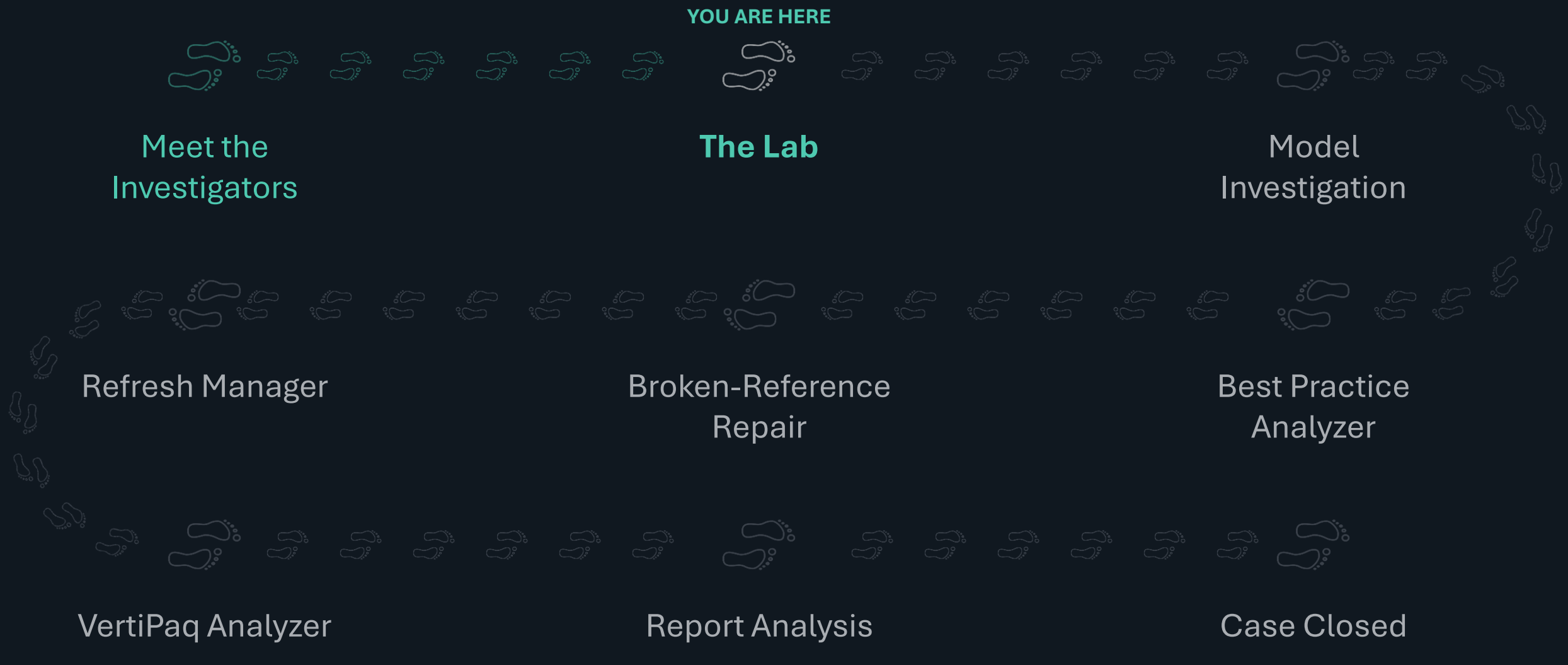
The investigation toolkit

Semantic Link Labs

- Created by Michael Kovalsky
- Built on Semantic Link
- BPA, VertiPaq, and report analysis tools
- Open source on GitHub
- Active development

Follow the Evidence

The Lab: Fabric notebooks



Fabric Notebook

Our investigation lab

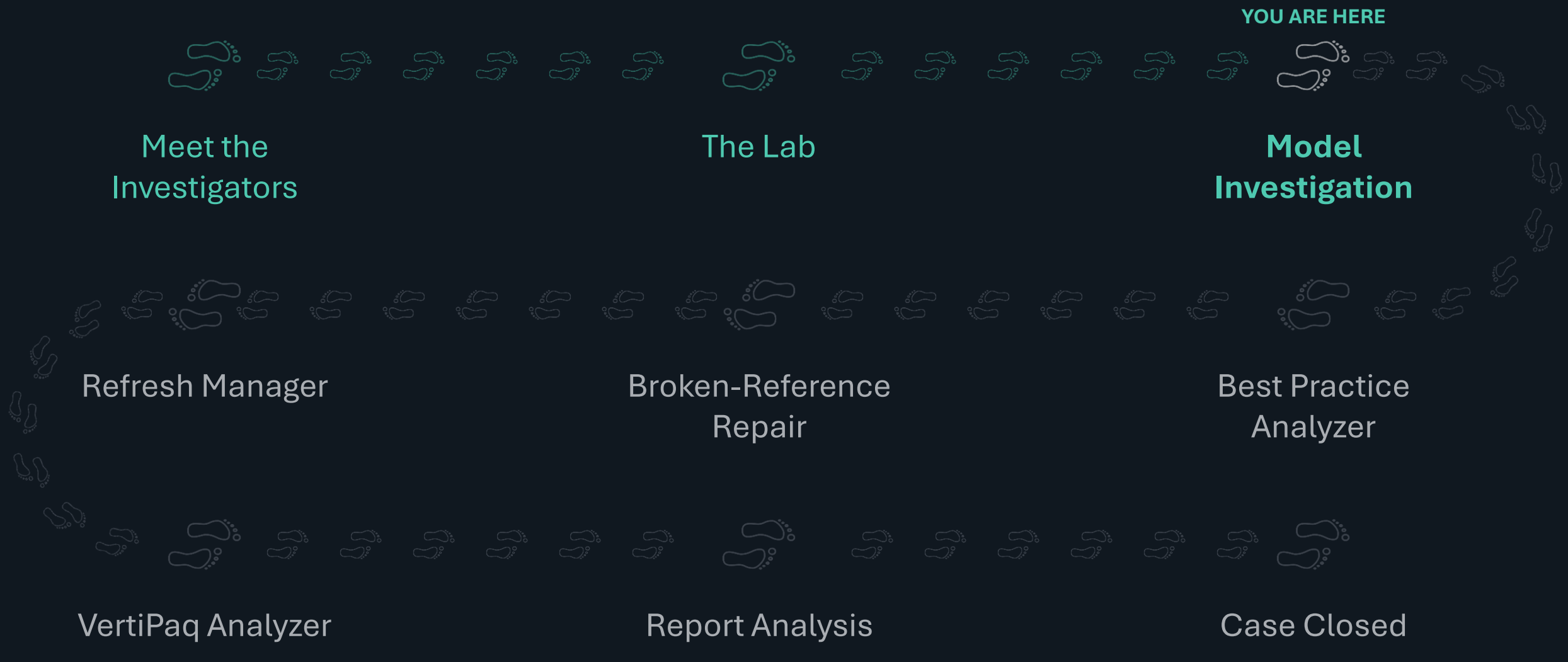


Semantic Link

Semantic Link Labs

Follow the Evidence

Model Investigation



Case File: The Unidentified Model

VICTIM

A semantic model you've inherited in an unfamiliar workspace.

SYMPTOMS

Tables, columns, and calculated columns need inspection.

Measure dependencies haven't been traced.

Case File: The Relationship Web

VICTIM

A semantic model with visible and hidden tables.

SYMPTOMS

The model shows the relationships we defined.

The data can reveal connections we haven’t defined yet.

Case File: Running the Numbers

VICTIM

DAX results we want to analyze in Python.

SYMPTOMS

We need DAX results in a DataFrame.

We need to filter those results using Python.

LIVE DEMO

Into the Lab

Model Investigation

Cases 1–3



Case Board: Investigations 1–3

Cases closed. Evidence on record.

CASE 01

The Unidentified Model

SOLVED

CASE 02

The Relationship Web

SOLVED

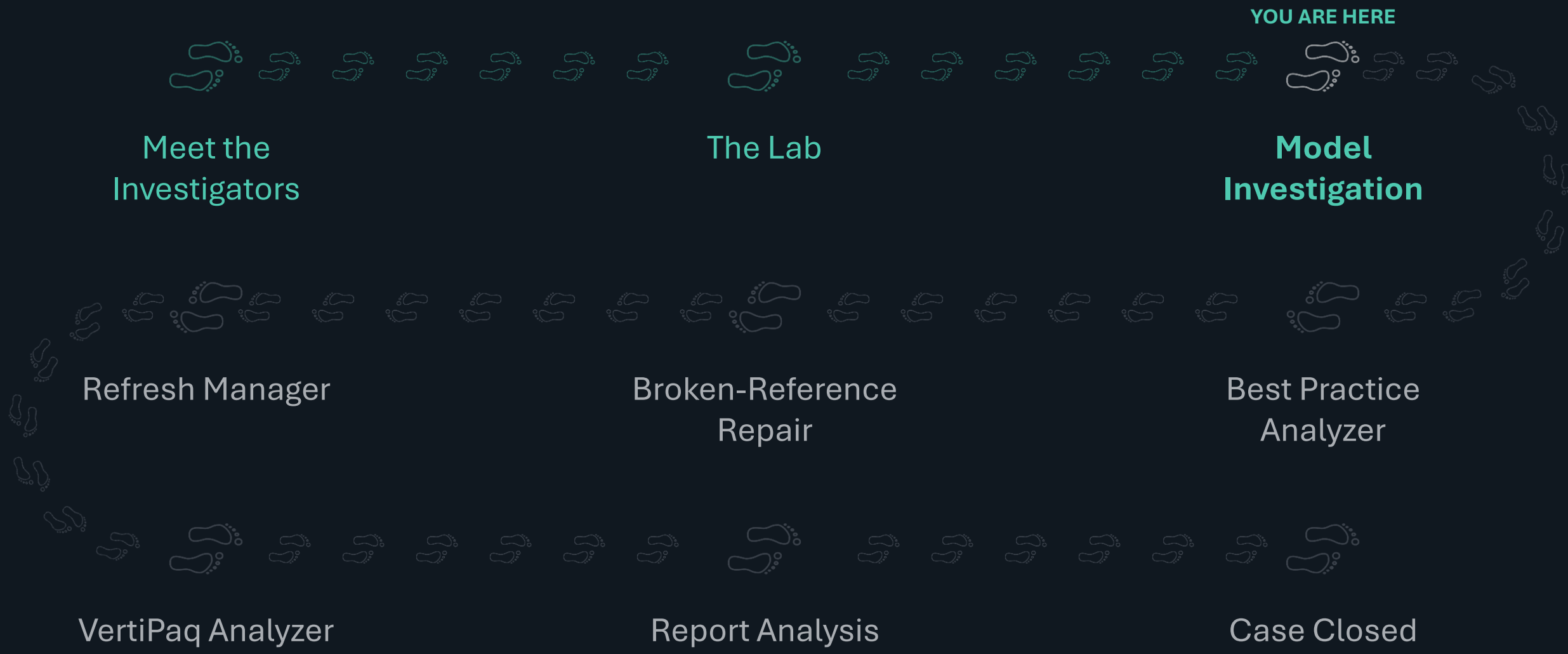
CASE 03

Running the Numbers

SOLVED

Follow the Evidence

Model Investigation · Cases 4–6



Case File: Something Doesn't Add Up

VICTIM

A semantic model with unmatched relationship keys.

SYMPTOMS

Some fact keys have no matching dimension row.

Blank members appear in slicers or visuals.

Case File: The Evidence Isn’t Ready Yet

VICTIM

A Delta table just written to OneLake.

SYMPTOMS

The write has completed in the notebook.

The table is still missing from the SQL endpoint.

Case File: The Witness Vanished

VICTIM

A Delta table deleted from the Lakehouse.

SYMPTOMS

The table was available through the SQL endpoint.

It is now missing, and its data needs recovery.

LIVE DEMO

Back to the Lab

Model Investigation

Cases 4–6



Case Board: Investigations 4–6

Cases closed. Evidence on record.

CASE 04

Something Doesn't Add Up

SOLVED

CASE 05

The Evidence Isn't Ready Yet

SOLVED

CASE 06

The Witness Vanished

SOLVED

Follow the Evidence

Best Practice Analyzer



Case File: The Workspace-Wide Sweep

VICTIM

Semantic models across the Fabric workspace.

SYMPTOMS

Checking one model leaves the others unexamined.

Quality issues accumulate between releases.

LIVE DEMO

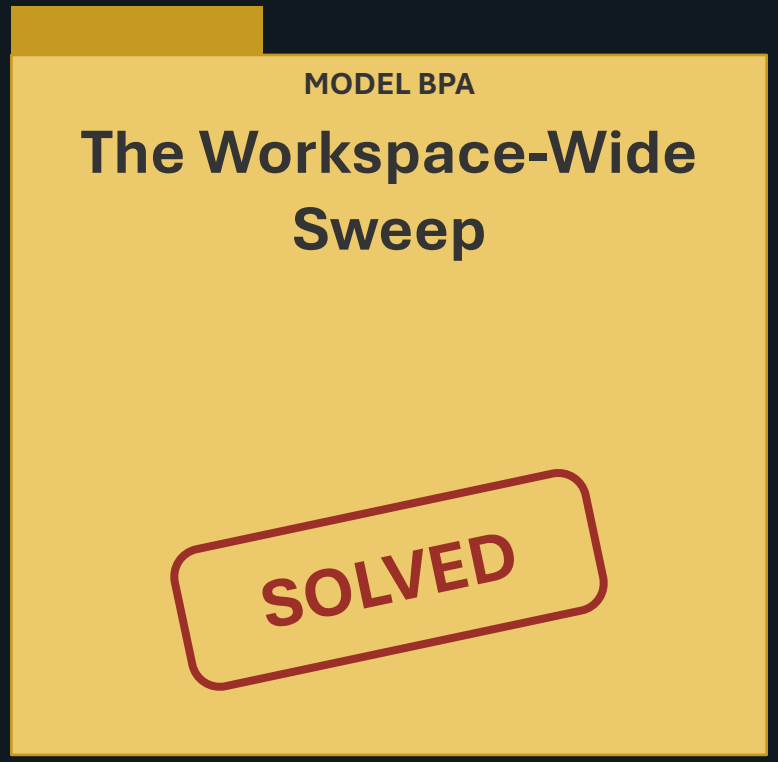
Back to the Lab

Best Practice Analyzer



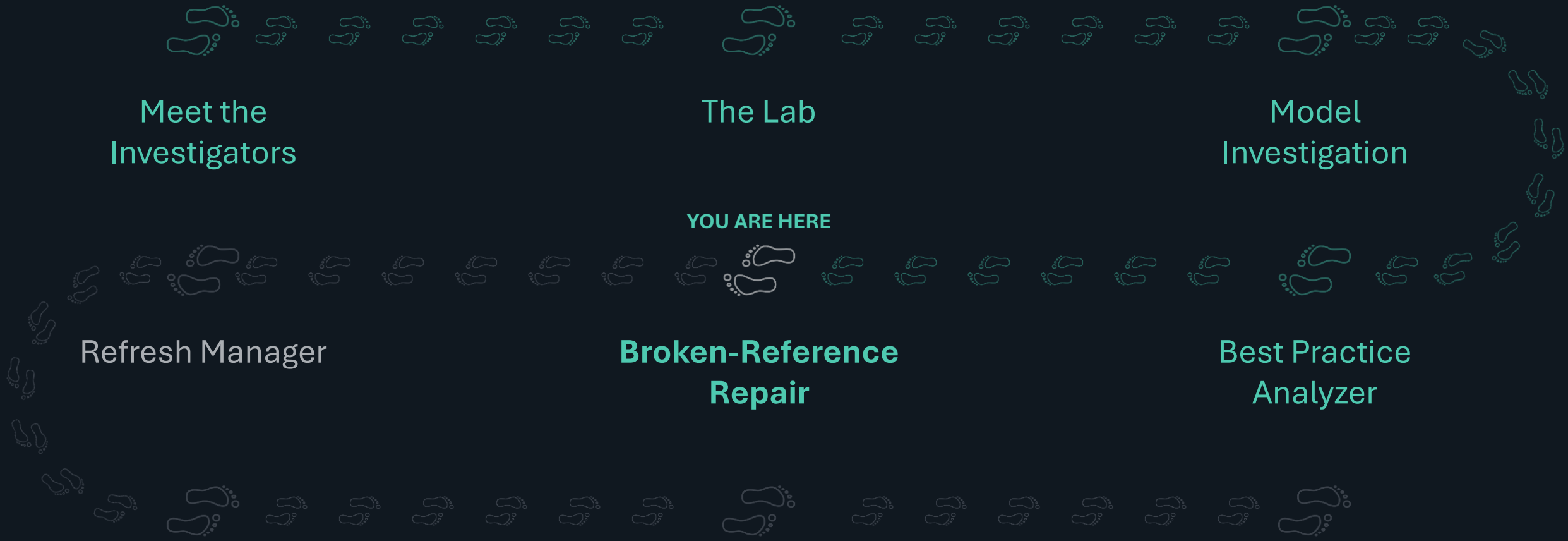
Case Closed: The Workspace-Wide Sweep

Issues identified. Results recorded. Report ready.



Follow the Evidence

Broken-Reference Repair



Case File: The Broken Reference

VICTIM

The SFCaseFiles_PNW_Lineage report.

SYMPTOMS

Incident Count no longer exists in the model.

The Incident Trend visual still references it and fails.

LIVE DEMO

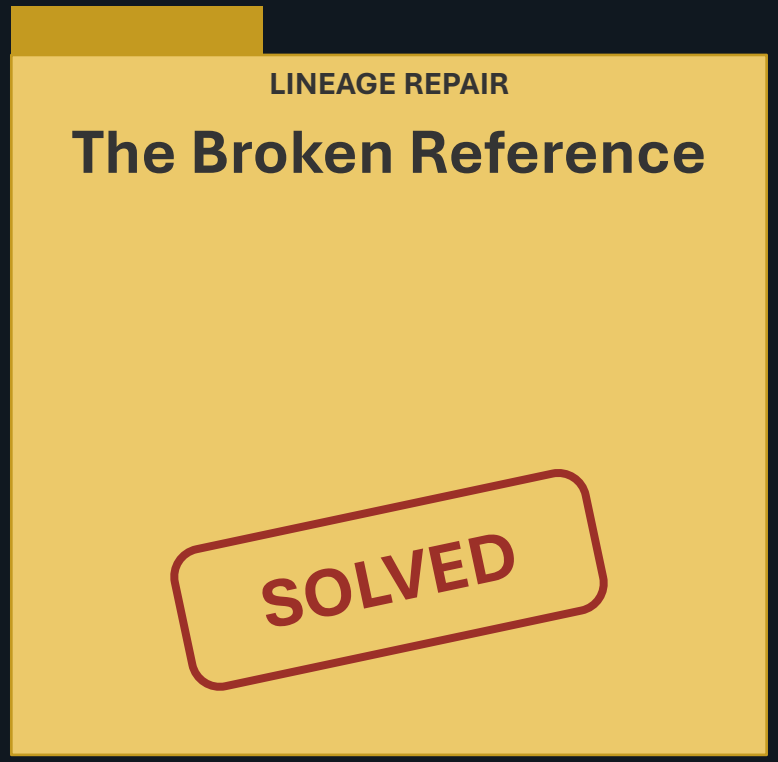
Back to the Lab

Broken-Reference Repair



Case Closed: The Broken Reference

Reference repaired. Visual verified.



Follow the Evidence

Refresh Manager



Meet the Investigators

The Lab

Model Investigation

YOU ARE HERE

Refresh Manager

Broken-Reference Repair

Best Practice Analyzer

VertiPaq Analyzer

Report Analysis

Case Closed

Case File: The Refresh Room

VICTIM

A semantic model whose refresh needs investigation.

SYMPTOMS

Refresh history needs inspection.

Refresh scope and schedule need review.

LIVE DEMO

Back to the Lab

Refresh Manager



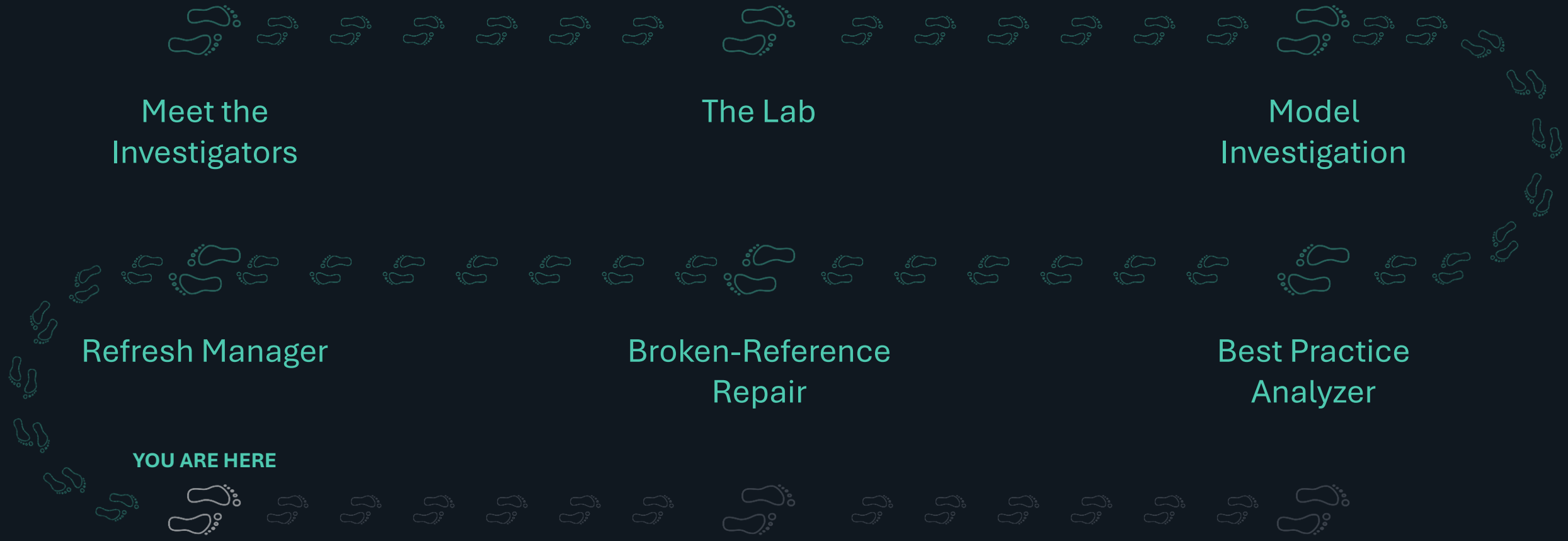
Case Closed: The Refresh Room

History inspected. Refresh controls and schedule understood.



Follow the Evidence

VertiPaq Analyzer



Case File: The Profile

VICTIM

Trudoso10K and Trudoso10K_WithAutoDateXL.

SYMPTOMS

Model size alone does not explain memory usage.

Hidden date tables also consume memory.

LIVE DEMO

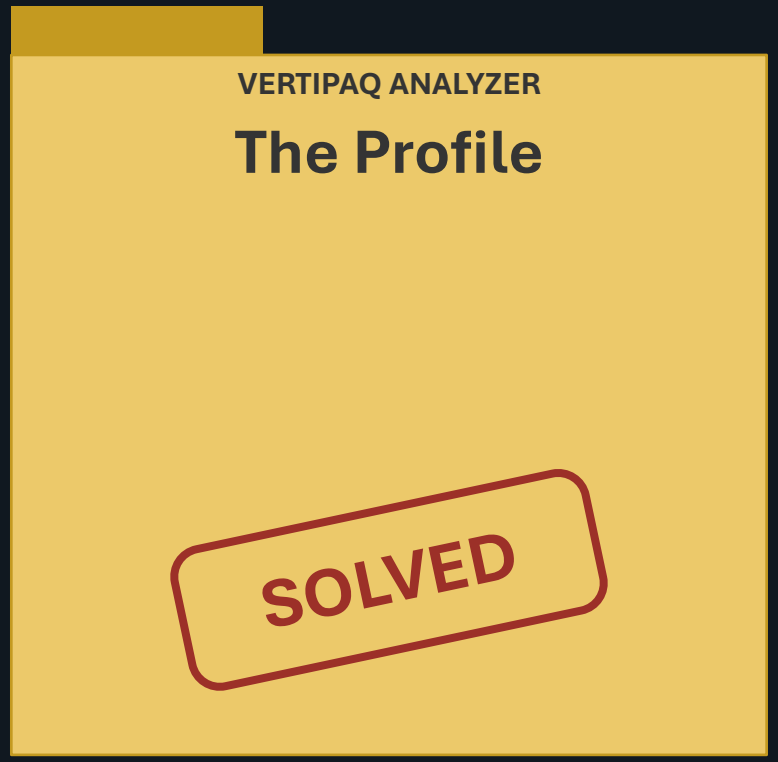
Back to the Lab

VertiPaq Analyzer



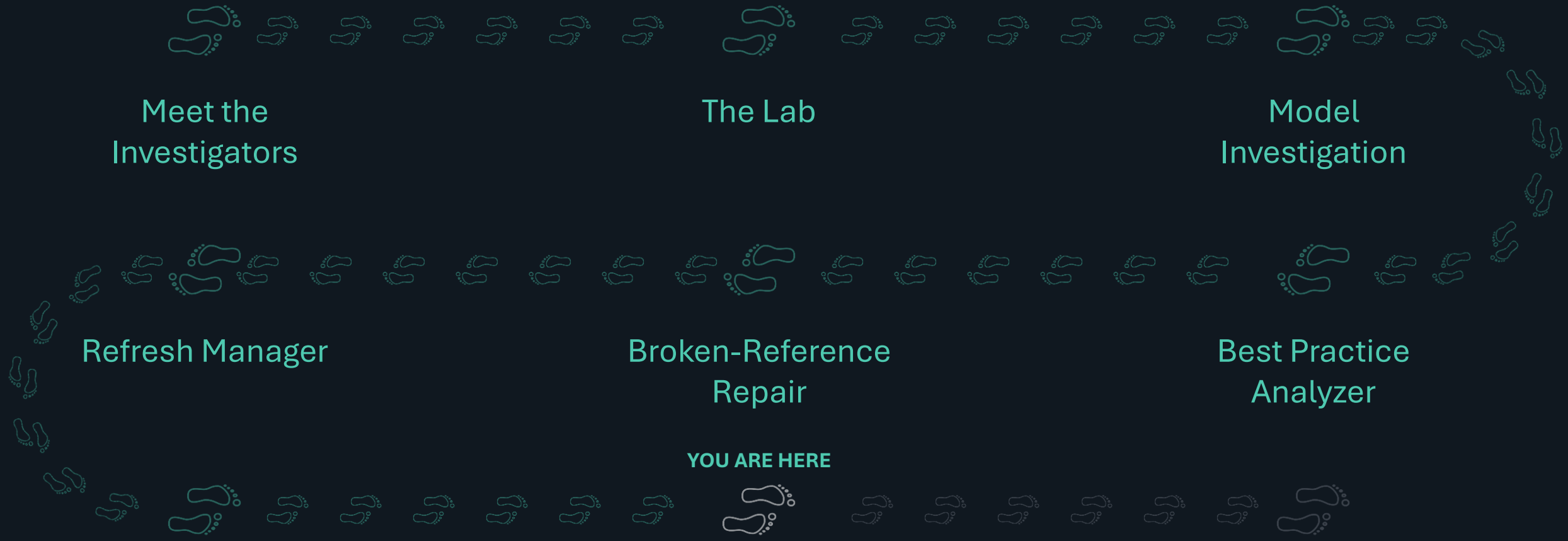
Case Closed: The Profile

Memory measured. Hidden date tables identified.



Follow the Evidence

Report Analysis



Case File: The Report That Lied

VICTIM

The SFCaseFiles_ReportLab_Thin report.

SYMPTOMS

Filters, bookmarks, and interactions affect the story.

Some calculations live in the report itself.

LIVE DEMO

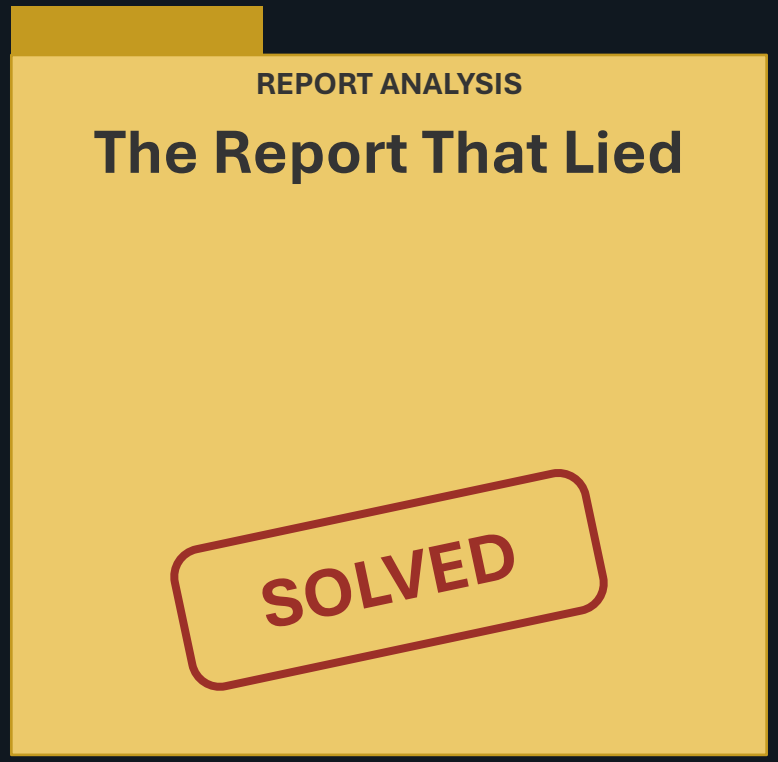
Back to the Lab

Report Analysis



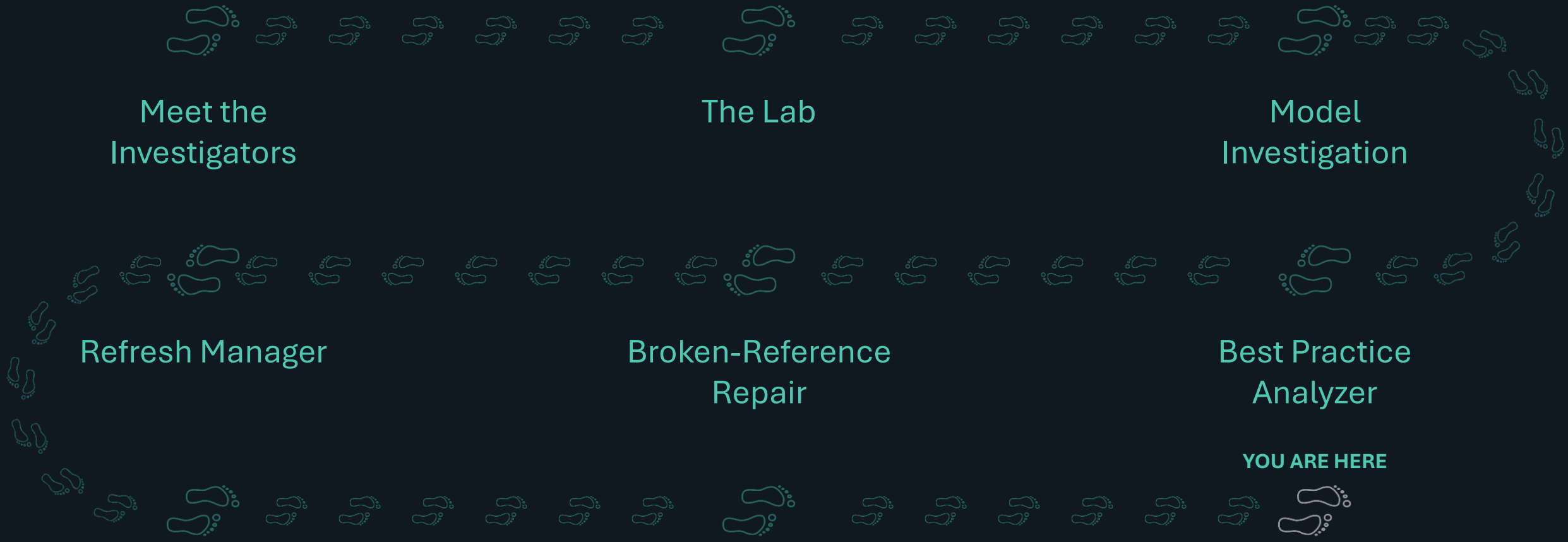
Case Closed: The Report That Lied

Filters, interactions, bookmarks, and report measures inspected.



Follow the Evidence

Case Closed: takeaways and next steps



Case Closed. What Will You Investigate?

Model quality?	Best Practice Analyzer
Refresh problems?	Refresh Manager
Model size?	VertiPaq Analyzer
Broken visuals or unexpected results?	Report and relationship analysis

Start with one question.
Use the notebook to follow the evidence.

The Detective’s Reference Desk

THE FIELD GUIDE

Semantic Link overview

How Power BI models connect to notebooks.

THE CASE FILES

Semantic Link Labs on GitHub

Code, examples, and project updates.

THE LAB MANUAL

Semantic Link Labs documentation

Function parameters and API reference.



THANK YOU

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